



BITS Pilani

Pilani|Dubai|Goa|Hyderabad

Introduction to Python for Data Science

DSECLPFDS / AIMLCPFDS

1) Introduction to *DSECLPFDS* / *AIMLCPFDS*

- Motivation & Objective
- Courseware, Canvas Walkthrough
- Books & Evaluation components
- Pedagogy for this course

2) Course Schedule

3) Getting started with Module 1

4) Q&A

Motivation for this course ?



Motivation

- As of now, Python is one of the most widely used programming languages in the Data Science field.
- Data Scientists just love Python!
- Python is easy to learn & has a great community for support!
- We would use Python for all the assignments / case-studies (For all the subjects in MTech DSE / MTech AIML).

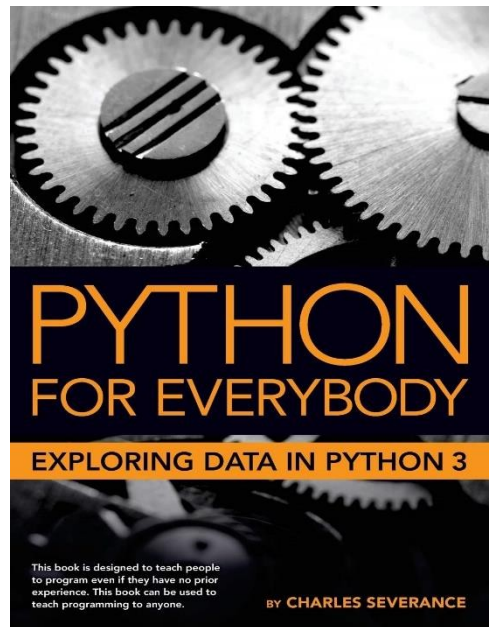
What is this course about ?

- Introduce the fundamental programming concepts of Python
- Enable you to solve data problems using Python
- Act as a kick-start / bridge for participants of the DSE/ AIML programme who are *new* to Python.

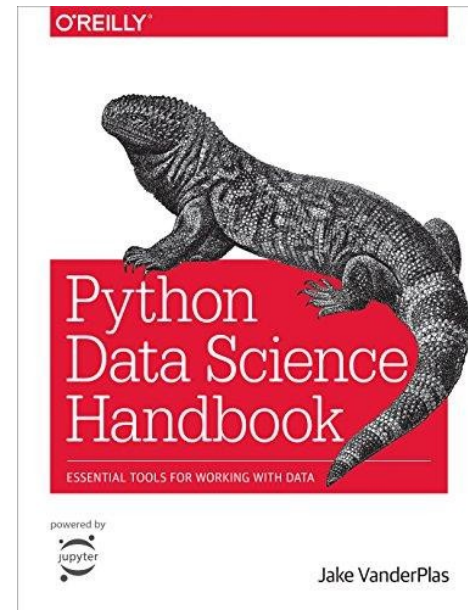
What is this course *not* about ?

- Comprehensive, in-depth discussion about Python programming.
- Comprehensive, in-depth discussion about data analysis using Python and related packages, libraries, and tools.

Text Books



Charles Severance: Python for Everybody,
Exploring Data in Python 3



Jake VanderPlas: Python Data Science
Handbook

eBooks of both are made available in Canvas (Course → Files → Books).
Do not publish these on the Internet. For our use, we have got permission from the authors to use the eBook.

Note: These are the prescribed ones. Please feel free to explore any Python materials that suits you.

Evaluation Components



- This course is **NOT evaluated** !!
- You will **not** have any exams for this course
- Nevertheless, there would be some exercises for you to try and hone your skills: **[No need to submit]**.
 - These will be on Canvas so you get accustomed to the Canvas LMS.
 - There will be “Assignments”
 - There will be “Quizzes”

Pedagogy for this Course



Step 01: Class Session

- **We** learn Fundamentals !
- Look at few examples for each concept.

Step 02: Explore

- **You** explore the additional notebooks.
- Practice more examples for each concept.

Step 03: Doubts

- Put your queries in Discussion Forum.
- Peers and TA to answer ...

Non-Beginners: You can directly start with Step 02 and also use this phase for additional learning which might help in future ... You play an important role in Step 3 as well in answering your peer's queries.

Program & Programming Language



Computer Program

- Set of instructions that perform a specific task executed by computer
- Required by computer to function
- Written by programmer using programming languages
 - Like C, C++, Java, Python etc.
- Executed with compiler and interpreter

Python as a Programming Language

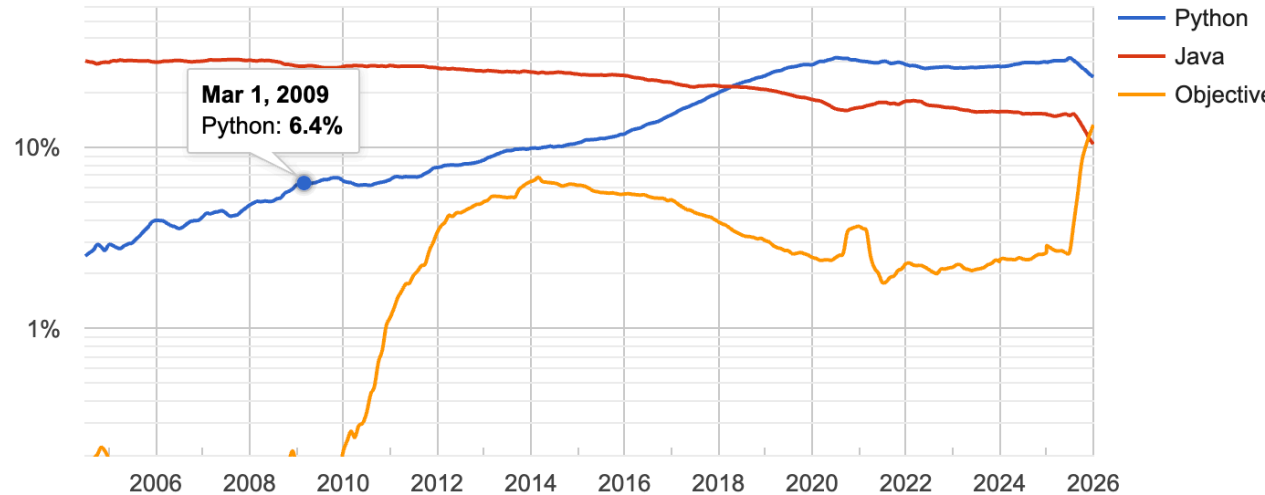


Why Python ?

Worldwide, Jan 2026 :

Rank	Change	Language	Share	1-year trend
1		Python	24.61 %	-5.0 %
2	↑↑	C/C++	14.13 %	+6.9 %
3	↑↑↑↑↑↑↑	Objective-C	13.35 %	+10.5 %
4	↓↓	Java	10.45 %	-4.8 %
5	↑	R	6.18 %	+1.6 %
6	↓↓↓	JavaScript	4.68 %	-3.3 %
7	↑↑↑↑	Swift	3.68 %	+1.1 %
8	↓	PHP	2.95 %	-0.9 %
9	↓↓↓↓	C#	2.79 %	-3.4 %
10	↑↑↑↑↑	Ada	2.71 %	+1.5 %

PYPL Popularity of Programming Language



Worldwide, Python is the most popular language ...
Source : <https://pypl.github.io/PYPL.html>

Python as a Programming Language



Python

- Designed by Guido van Rossum around 1990
- Not just a scripting language
- Easy to learn, read, use
- Extensible (add new modules)
- Highly readable
- Latest Version 3.13.x
- Most fond of language for Data Scientists

Properties

- Open Source
 - copyrighted but use not restricted
 - owned by independent non-profit, PSF
- Mature (30+ years old)
- Supportive user community
 - plenty of good books, too
 - Active user community
- Simple design, easy to learn
 - reads like “pseudo-code”
 - Suitable as first language

Use Python for...

Web Development: Django , Pyramid , Bottle , Tornado , Flask , web2py

GUI Development: tkinter , PyGObject , PyQt , PySide , Kivy , wxPython

Scientific and Numeric: SciPy , Pandas , IPython

Software Development: Buildbot , Trac , Roundup

System Administration: Ansible , Salt , OpenStack

Components of Python World :

- Core Python
- Distributions
- Frameworks / IDEs
- Third party Libraries

Core Python

- Programming Language itself
- Some standard modules are available
- Other packages needs to be explicitly installed

Python Distribution

- Python + packages
- Majority of packages, libraries are already available
- Package management is simplified
 - Anaconda from Continuum Analytics
 - IPython and its IPyKit variant

Frameworks / IDEs

- Use frameworks to create code and develop applications
- Provides a defined structure to the developers so that they can focus on the core logic of the application rather than on other elements
- Python web framework
 - ✓ Django
 - ✓ Web2py
 - ✓ Flask
- Python IDEs
 - ✓ IDLE
 - ✓ PyCharm
 - ✓ Spyder
- ✓ Jupyter Notebooks

Third party Libraries

- Makes life of developers very simple
- Just need to know the right library to carry out a task
 - NumPy
 - Scipy
 - Pandas
 - Matplotlib
 - Seaborn
 - Bokeh
 - ScikitLearn
 - And List goes on ...

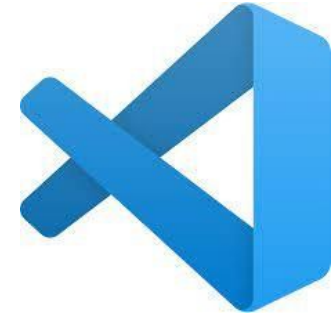
Three Ways :

- Install Python directly
 - Install the Python language with installer
 - Need to install other packages explicitly using pip install
 - <https://www.python.org/downloads/>
- Use Python distribution
 - The open-source Anaconda Distribution is the easiest way to perform Python coding
 - Works on Linux, Windows, and Mac OS X
 - <https://docs.anaconda.com/anaconda/install/windows/>
- Use Cloud based services
 - The simplest of all but needs internet connectivity to use
 - Microsoft Azure Notebooks
 - Google Collab

Integrated Development Environments (IDE) for Python



Common IDE's:

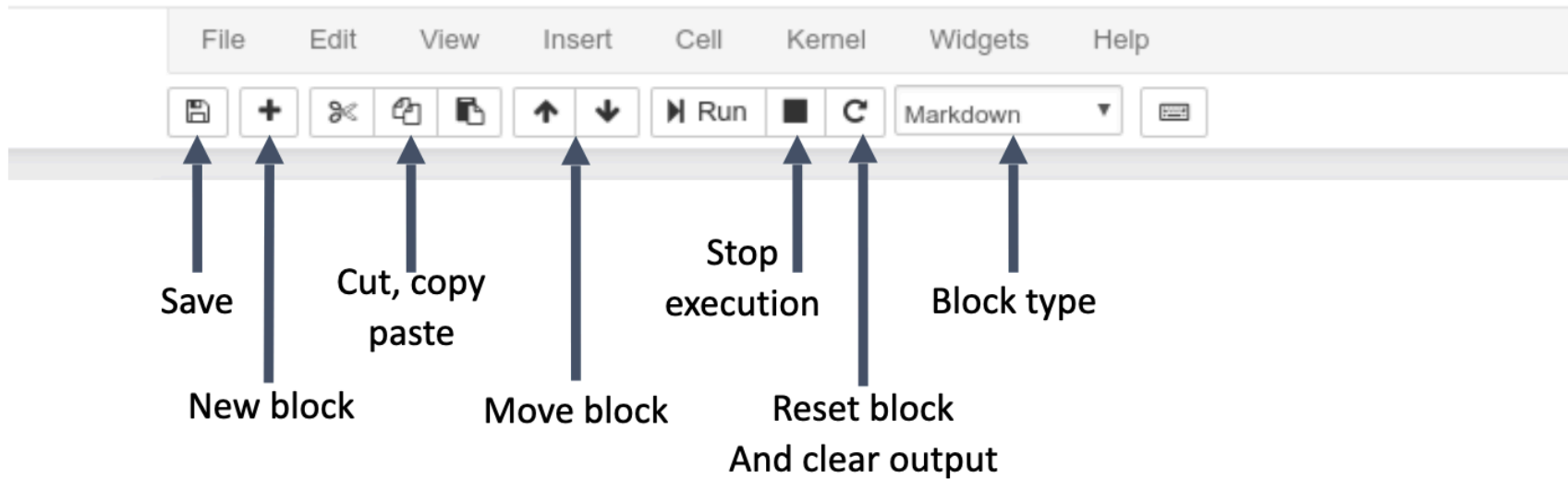


Our Favourite (For MTech Programme):



- **.py** is a regular python file. It's plain text and contains just your code.
- **.ipynb** is a python notebook and it contains the notebook code, the execution results and other internal settings in a specific format.

Tool Bar

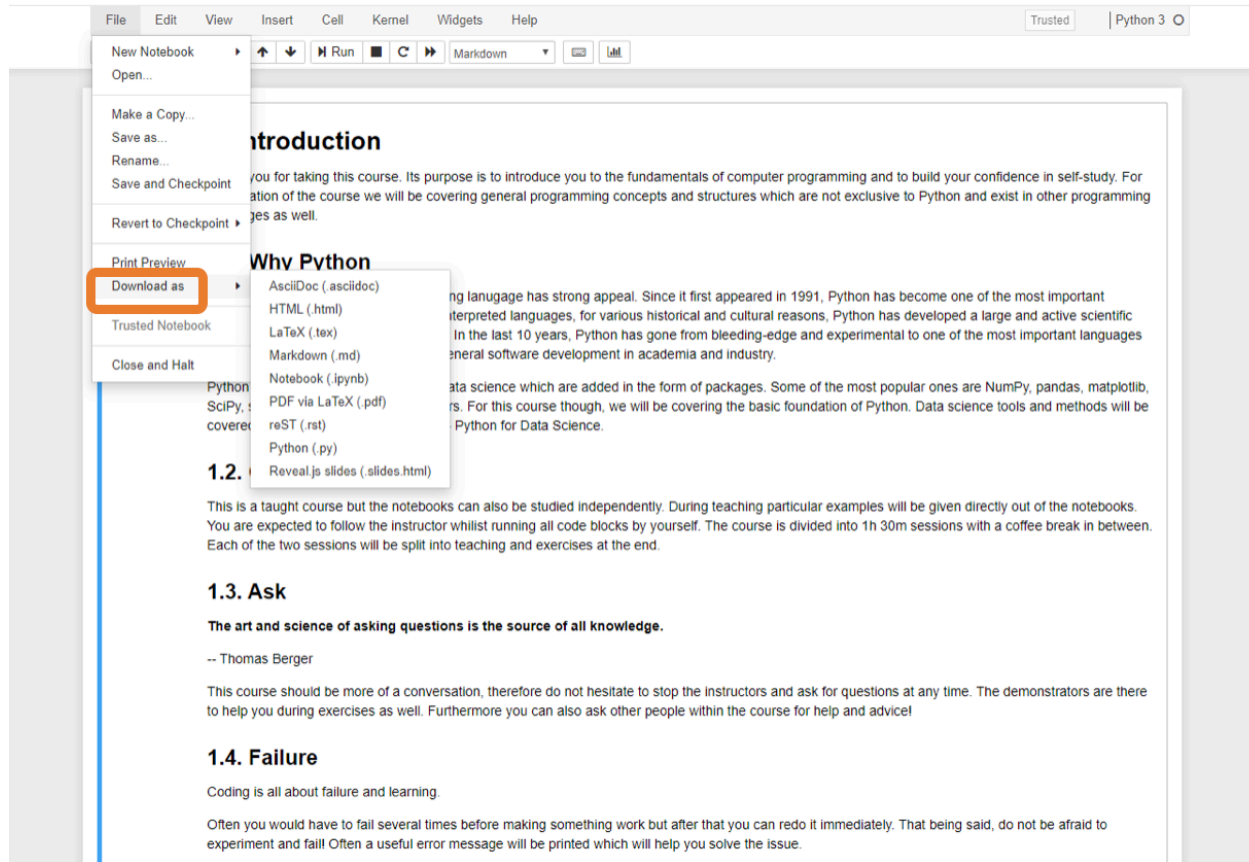


Download Files

innovate

achieve

lead



The screenshot shows a Jupyter Notebook interface. The 'File' menu is open, and the 'Download as' option is highlighted with an orange box. The 'Download as' submenu is also visible, showing various file formats for downloading the notebook: AsciiDoc (.asciidoc), HTML (.html), LaTeX (.tex), Markdown (.md), Notebook (.ipynb), PDF via LaTeX (.pdf), reST (.rst), Python (.py), and Reveal.js slides (.slides.html). The background content of the notebook is visible, showing sections titled 'Introduction', 'Why Python', '1.2.', '1.3. Ask', and '1.4. Failure'.

Introduction

you for taking this course. Its purpose is to introduce you to the fundamentals of computer programming and to build your confidence in self-study. For ation of the course we will be covering general programming concepts and structures which are not exclusive to Python and exist in other programming es as well.

Why Python

ng lanugage has strong appeal. Since it first appeared in 1991, Python has become one of the most important interpreted languages, for various historical and cultural reasons, Python has developed a large and active scientific In the last 10 years, Python has gone from bleeding-edge and experimental to one of the most important languages general software development in academia and industry.

ata science which are added in the form of packages. Some of the most popular ones are NumPy, pandas, matplotlib, rs. For this course though, we will be covering the basic foundation of Python. Data science tools and methods will be Python for Data Science.

1.2.

This is a taught course but the notebooks can also be studied independently. During teaching particular examples will be given directly out of the notebooks. You are expected to follow the instructor whilst running all code blocks by yourself. The course is divided into 1h 30m sessions with a coffee break in between. Each of the two sessions will be split into teaching and exercises at the end.

1.3. Ask

The art and science of asking questions is the source of all knowledge.

-- Thomas Berger

This course should be more of a conversation, therefore do not hesitate to stop the instructors and ask for questions at any time. The demonstrators are there to help you during exercises as well. Furthermore you can also ask other people within the course for help and advice!

1.4. Failure

Coding is all about failure and learning

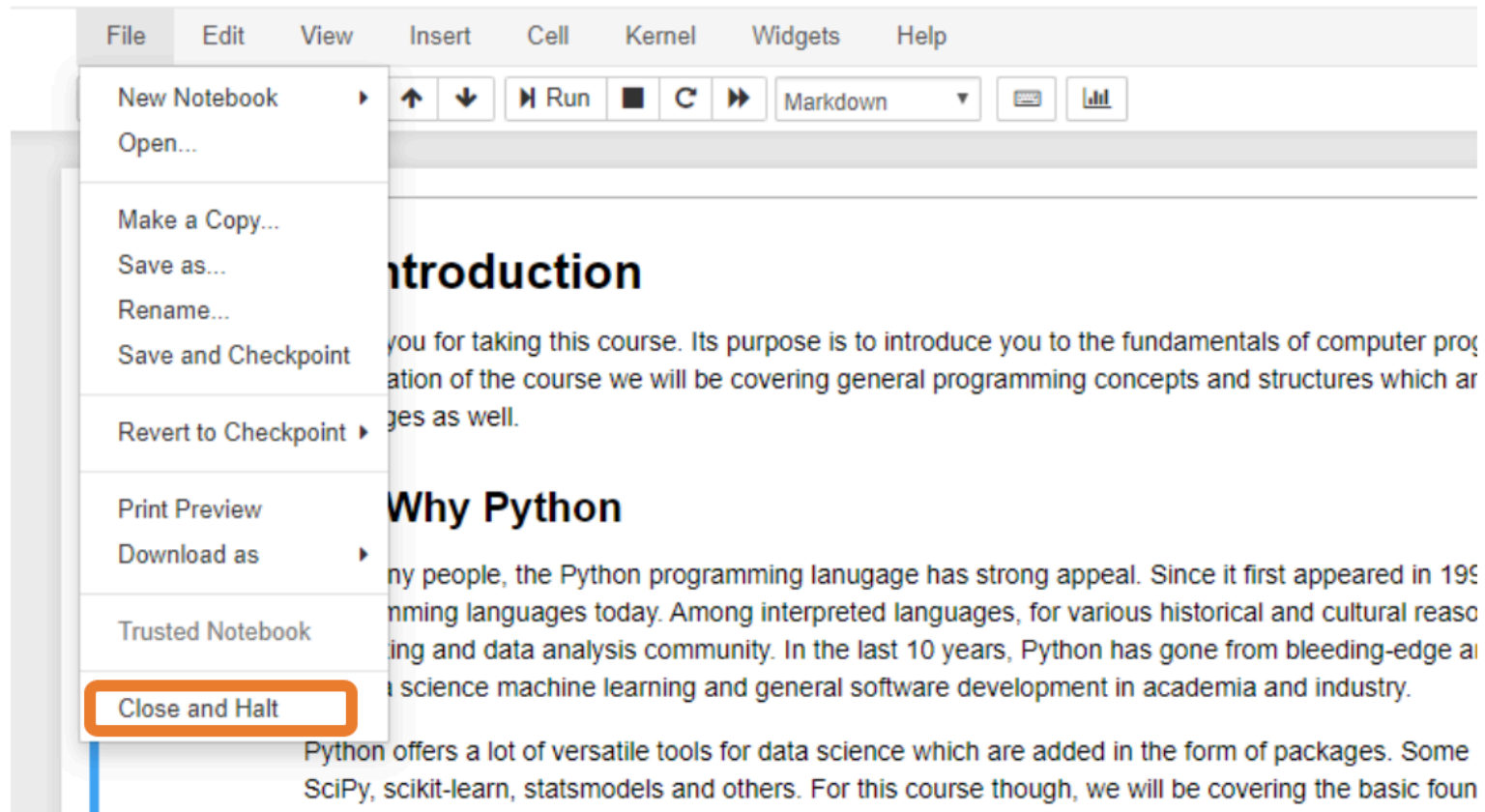
Often you would have to fail several times before making something work but after that you can redo it immediately. That being said, do not be afraid to experiment and fail! Often a useful error message will be printed which will help you solve the issue.

Edit / Delete Cells



The screenshot displays the Jupyter Notebook application interface. The top menu bar includes 'File', 'Edit', 'View', 'Insert', 'Cell', 'Kernel', 'Widgets', and 'Help'. The 'Edit' menu is open, showing various options. The 'Delete Cells' option is highlighted with a red rectangle. Other options in the 'Edit' menu include 'Cut Cells', 'Copy Cells', 'Paste Cells Above', 'Paste Cells Below', 'Paste Cells & Replace', 'Undo Delete Cells', 'Split Cell', 'Merge Cell Above', 'Merge Cell Below', 'Move Cell Up', 'Move Cell Down', 'Edit Notebook Metadata', 'Find and Replace', 'Cut Cell Attachments', 'Copy Cell Attachments', 'Paste Cell Attachments', and 'Insert Image'. The background shows a notebook cell with text about a course, including sections like 'Introduction' and 'Python'.

File Close/ Halt



Input / Output with Python



- `print()` can be used to output a message
- `input()` can be used to enter an input to the python program.
- `#` can be used to provide comments.
- `"""` (triple quotes) can be used to write documentation.

Demo:

- Let's see how to launch Jupyter Notebook
- See the basics of Notebook
- Practice some I/O statements and comments.

Variables actually have a type, which defines the way it is stored.

The basic types are:

Type	Declaration	Example	Usage
Integer	int	<code>x = 124</code>	Numbers without decimal point
Float	float	<code>x = 124.56</code>	Numbers with decimal point
String	str	<code>x = "Hello world"</code>	Used for text
Boolean	bool	<code>x = True</code> or <code>x = False</code>	Used for conditional statements
NoneType	None	<code>x = None</code>	Whenever you want an empty variable

Python offers us a way of converting variables to different types!

Casting – the operation of converting a variable to a different type

```
x = 10      # This is an integer
y = "20"    # This is a string
x + int(y)
```

30

Arithmetic Operations



Similar to actual Mathematics.

Order of precedence is the same as in Mathematics.

We can also use parenthesis ()

Symbol	Task Performed	Example	Result
+	Addition	$4 + 3$	7
-	Subtraction	$4 - 3$	1
/	Division	$7 / 2$	3.5
%	Mod	$7 \% 2$	1
*	Multiplication	$4 * 3$	12
//	Floor division	$7 // 2$	3
**	Power of	$7 ** 2$	49

Comparison Operators



- Return Boolean values
(i.e. True or False)
- Used extensively for conditional statements

Operator	Output
$x == y$	True if x and y have the same value
$x != y$	True if x and y don't have the same value
$x < y$	True if x is less than y
$x > y$	True if x is more than y
$x \leq y$	True if x is less than or equal to y
$x \geq y$	True if x is more than or equal to y

Strings



- Powerful and flexible in Python
- Can be added
- Can be multiplied
- Can be multiple lines

Strings



```
x = "Python"  
y = "rocks"  
x + " " + y
```

'Python rocks'

```
x = "This can be"  
y = "repeated "  
x + " " + y * 3
```

'This can be repeated repeated repeated '

Strings



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'Python rocks'

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x + " " + y * 3
```

'This can be repeated repeated repeated '

Strings and Numbers



Often we would need to mix up numbers and strings. It is best to keep numbers as numbers (i.e. int or float) and cast them to strings whenever we need them as a string.

```
x = 6
x = ( x * 5345 ) // 63
"The answer to Life, the Universe and Everything is " + str(x)
'The answer to Life, the Universe and Everything is 42'
```

Strings and Numbers



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